Black Board Talk for 2023-11-08

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Reference: Klessen and Glover

Euler eqn: $(\nabla \cdot u)\, u$ is convective/advectiwve acceleration. Describes fluid's velocity changes along the flow.

1 Cosmology

Friedmann eqn:

$$\frac{H^2}{H_0^2} = \Omega_{r,0} \left(\frac{a_0}{a}\right)^4 + \Omega_{m,0} \left(\frac{a_0}{a}\right)^3 + \Omega_{k,0} \left(\frac{a_0}{a}\right)^2 + \Omega_{\Lambda,0}$$

With $\Omega_i(t) = \frac{8\pi G}{3H^2}\rho_i(t)$ It can be written in more detail.

Remember that in comoving coordinates:

$$\nabla_x = a^{-1}\nabla_r$$

$$(\partial_t)_x = (\partial_t)_q - (v_0 \cdot \nabla_x)$$

Since

$$\left(\frac{\partial f(\mathbf{x} = a\mathbf{q}, t)}{\partial t}\right)_{\mathbf{q}} = \left(\frac{\partial f}{\partial t}\right)_{\mathbf{x}} + \dot{a}q^i \left(\frac{\partial f}{\partial x^i}\right)_t$$

$x(t) = \frac{r(t)}{a(t)}$

Gravitational potential perturbation:

$$\phi(x,t) = \Phi(r,t) - \frac{1}{2}a\ddot{a}x^2$$

The second term here is the background potential, which we get after Taylor expansion. The 0^{th}-order term of the Taylor is just constant potential, which can be subtracted w/o loss of generality.

The 1^{st}-order term of the Taylor would correspond to force, which corresponds to non-homogeneous and non-isotropic Universe.

The 2^{nd}-order term corresponds to tidal forces.

It comes from taking both Poisson and the 2^{nd} Friedmann eqn:

$$\begin{aligned} a^{-2}\nabla_x^2\Phi &= 4\pi G\rho \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}\left(\rho + 3P\right) \end{aligned} \tag{1}$$

Plug in ρ from Friedmann to Poisson, and integrate w.r.t. x (a is const in x)

Density perturbation:

$$\delta(x,t) = \frac{\rho(x,t) - \langle\rho\rangle(t)}{\langle\rho\rangle(t)}$$

Velocity perturbations:

$$u = \frac{\mathrm{d}r}{\mathrm{d}t} = \dot{a}x + a\dot{x} = v_H(x,t) + a(t)\dot{x}$$

So peculiar velocity is our perturbation: $v = a(t)\dot{x}$

2 Jeans eqns in comoving frame

This is how we actually obtain the Jean's equation for comoving frame:

Start with the fluid equations (Continuity, Euler and Poisson):

$$\begin{aligned} \frac{\partial\rho}{\partial t} + \vec{\nabla}_r\cdot\rho\vec{u} &= 0 \\ \frac{\partial\vec{u}}{\partial t} + \left(\vec{u}\cdot\vec{\nabla}_r\right)\vec{u} &= -\frac{1}{\rho}\vec{\nabla}_r P - \vec{\nabla}_r\Phi \\ \nabla_r^2\Phi &= 4\pi G\left(\rho + \frac{3P}{c^2}\right) \end{aligned}$$

Here we drop the pressure terms in the 3^{rd} eqn, since we're looking at the Universe after the recombination (there that pressure term corresponds to radiation pressure). The pressure term in eqn. 3 corresponds to the gas pressure.

We take the pressure perturbation as $p = c_s^2\delta$, since $c_s^2 = \left(\frac{\partial P}{\partial\rho}\right)_s$

$$\begin{aligned} \frac{\partial\rho}{\partial t} + \vec{\nabla}_r\cdot\rho\vec{u} &= 0 \\ \frac{\partial\vec{u}}{\partial t} + \left(\vec{u}\cdot\vec{\nabla}_r\right)\vec{u} &= -\vec{\nabla}_r\Phi - c_s^2\vec{\nabla}^2\delta \\ \nabla_r^2\Phi &= 4\pi G\rho \end{aligned}$$

We then transform it into the equations for the comoving coordinates (in terms of perturbations):

$$
\begin{aligned}
\frac{\partial \delta}{\partial t} + \frac{1}{a}\vec{\nabla}_x \cdot (1+\delta)\vec{v} = 0 \\
\frac{\partial \vec{v}}{\partial t} + \frac{1}{a}\left(\vec{v}\cdot\vec{\nabla}_x\right)\vec{v} + \frac{\dot{a}}{a}\vec{v} = -\frac{1}{a}\vec{\nabla}_x\phi \\
\nabla_x^2\phi = 4\pi G a^2 \rho_u \delta
\end{aligned}
$$

Taking linear approximation (small perturbations in velocity and density):

$$
\begin{aligned}
\frac{\partial \delta}{\partial t} + \frac{1}{a}\vec{\nabla}\cdot\vec{v} = 0 \\
\frac{\partial \vec{v}}{\partial t} + \frac{\dot{a}}{a}\vec{v} = -\frac{1}{a}\vec{\nabla}\phi \\
\nabla^2\phi = 4\pi G a^2 \rho_u \delta
\end{aligned}
$$

These are: continuity equation, Euler equation, Poisson equation.
Then we take divergence of the Euler equation, to get:

$$
\frac{\partial}{\partial t}\left(\vec{\nabla}\cdot\vec{v}\right) + \frac{\dot{a}}{a}\left(\vec{\nabla}\cdot\vec{v}\right) = -\frac{1}{a}\vec{\nabla}^2\phi - c_s^2\vec{\nabla}^2\delta
$$

Now we use the expression for $\vec{\nabla}\cdot\vec{v}$ from continuity eqn, and the expression for $\nabla^2\phi$ from the Poisson eqn, to finally obtain:

$$
\frac{\partial^2 \delta}{\partial t^2} - c_s^2\vec{\nabla}^2\delta + 2H\frac{\partial \delta}{\partial t} = 4\pi G\rho_u\delta \tag{2}
$$

Solution to this eqn (for non-comoving coords):

$$
\delta = \delta_0 e^{i(\omega t - kx)}
$$

With:
Jeans length:

$$
\lambda_J = \left(\frac{15k_B T}{4\pi G\mu\rho}\right)^{\frac{1}{2}}
$$

3 Thermodynamics

The problem of formation of Pop III stars is mostly thermodynamical

$$
\gamma_{eff} = 1 + \frac{\mathrm{d}\ln T}{\mathrm{d}\ln p}
$$

Formation pathways for H_2:

$$
\begin{aligned}
H + e^- \rightarrow H^- + \gamma \\
H^- + H \rightarrow H_2 + e^-
\end{aligned} \tag{3}
$$

$$
\begin{aligned}
H + H^+ &\rightarrow H_2^+ + \gamma \\
H_2^+ + H &\rightarrow H_2 + H^+
\end{aligned} \tag{4}
$$

Slow, since formation rates of H_2^+ and H^- are low.

For high particle densities, above $n \sim 10^9$ cm^{-3}, another process becomes important, the three-body reaction:

$$H + H + H \rightarrow H_2 + H \tag{5}$$

All atomic hydrogen is converted to H_2 once the densities of 10^{11} are reached. Above $n \sim 10^{13}$ the H_2 becomes collisionally dissociated, and the gas eventually turns atomic again.

At high redshifts, scattering of cosmic microwave background photons off electrons can effectively cool the gas.

Collisional-Induced Excitation (CIE) is when H_2 molecules induce dipole moment when they are in proximity of each other, allowing dipole transitions. That is important since ro-vibrational transitions in H_2 are generally forbidden as dipole transitions. They can happen as quadrupoles though.

Heating	**Cooling**	
H_2 formation	H_2 ro-vibrational	H_2
	H_2 CIE	
	HD	
PdV heat	H_2 dissociation	
	He^3	
	LiH	
	Compton atomic H	

Table 1: Your caption here

4 Collapse

The labels in the $\log_{10} T - \log_{10} n$ plot indicate key phases of the initial collapse:

(A) As the gas begins to flow into the potential well of the dark matter halo, it is compressionally heated to the virial temperature of the system. Isothermal-like collapse!

(B) At the same time the H_2 fraction increases from $< 10^{-5}$ to $\sim 10^{-3}$, which is sufficient for the gas to go into a run-away cooling phase

(C) and brings it down to the minimum temperature of ~ 200 K (C).

- As more gas flows into the halo center, eventually the potential becomes dominated by the gas rather than by dark matter.

- The contraction proceeds and the heat provided by PdV work begins to dominate again over cooling.

(D) As a consequence, the gas temperature rises to about 1000 K at $n \sim 10^9$ cm^{-3}

(E) At this stage, three-body H_2 formation (Equation 17) becomes important and the gas quickly becomes fully molecular

- Below densities $n \sim 10^{13}$ cm^{-3} the main cooling process is the (forbidden) ro-vibrational line emission from H_2.

(F) However, at larger n two additional cooling processes become important.

- The first is collision-induced emission (CIE), which occurs when two molecules come close to each other.
- Van der Waals forces can then induce a temporary dipole which allows for efficient dipole emission during the interaction time interval .
- The second one is associated with the collisional dissociation of H_2, which sets in at temperatures around 2000 K and quickly dominates the overall cooling behavior.

(G) As the density increases further (G), more and more H_2 molecules are destroyed.

- This also implies that more hydrogen atoms become available again for H_2 formation through the three-body process and the associated energy release starts to dominate the overall heating rate over PdV work.
- This continues until the molecular gas is largely adiabatic at $n \sim 10^{20}$ cm^{-3}.

EOS:

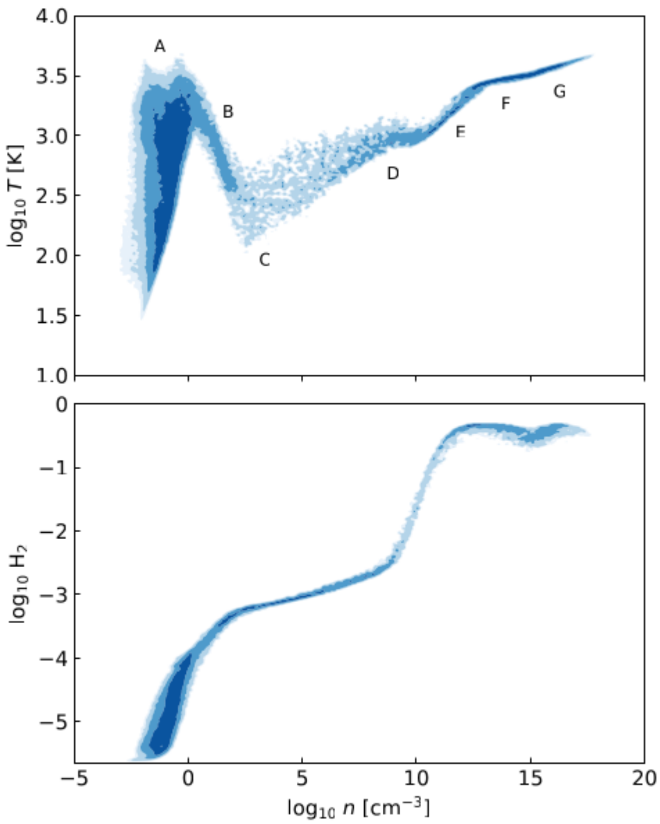

Figure 1

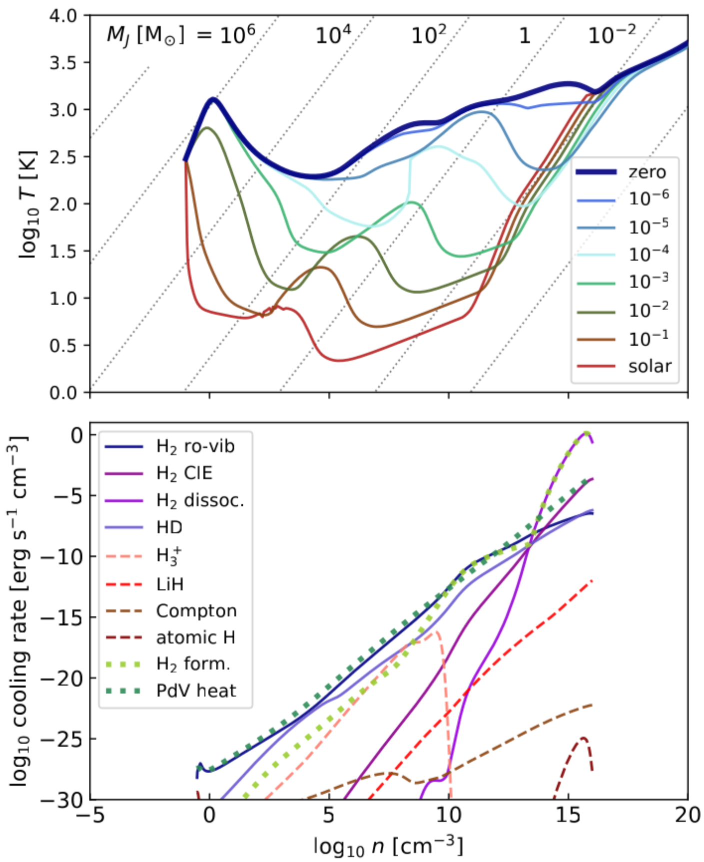

Figure 2

5 Accretion

Toomre (1964) instability criterion for infinitely thin disks:

$$Q = \frac{\kappa c_s}{\pi G \Sigma} \lesssim 1$$

κ is epicyclic frequency ($= \Omega$ for Keplerian rotation), Σ is surface density.

This instability occurs in differentially rotating self-gravitating thin disks if gravity dominates over pressure and rotational shear

6 Fragmentation

The reason for the high susceptibility of primordial accretion disks to fragmentation is always the same: For typical halo conditions in the primordial Universe the mass load onto the disk from the infalling envelope exceeds its capability to transport material inwards by gravitational or magnetoviscous torques. As a consequence massive spiral arms build up and speed up the inward transport. Often this is not enough, and the arms become non-linear and interact with each other, leading to run-away collapse in the interaction regions.

$$\text{Mass accretion rate, } \dot{M} = \int_{disk} \Sigma \mathbf{v} \cdot d\mathbf{A}$$

$$\text{Toomre's stability criterion, } Q = \frac{c_s \kappa}{\pi G \Sigma} < 1 \text{ (for fragmentation)}$$

$$\text{Gravitational torques, } \Gamma = \int_{disk} r^2 \Sigma \frac{\partial \Phi}{\partial r} d\mathbf{A}$$

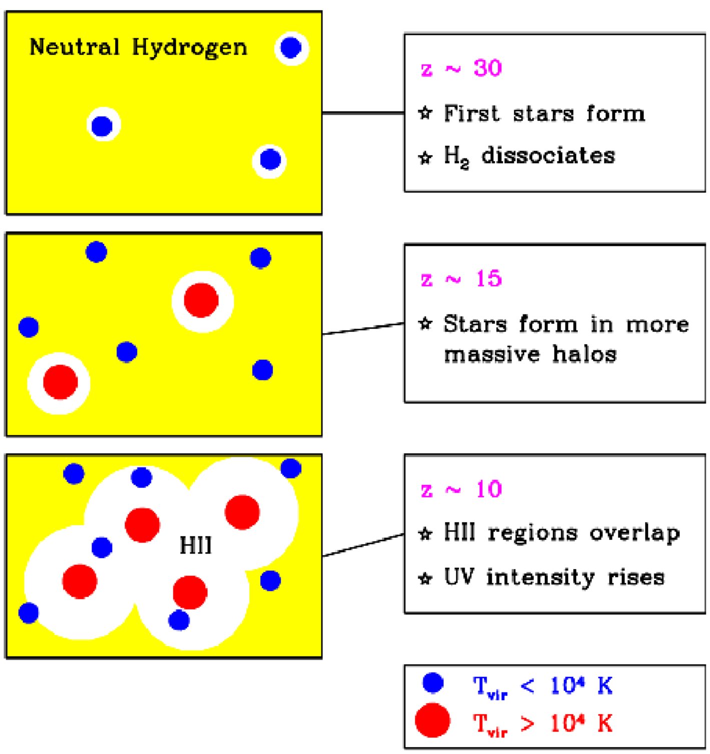

Figure 3

7 Few massive or many low-mass?

Cooling regime: $\gamma_{eff} < 1$

Heating regime: $\gamma_{eff} > 1$

Turbulent gas in cooling regime is predisposed to formation of many low-mass stars.

In heating regime, it forms few high-mass objects.

Transition between these regimes provides ideal conditions for efficient fragmentation and introduces characteristic mass scale to the system, which is $\approx$ Jeans mass.

At high redshifts, scattering of cosmic microwave background photons off electrons can effectively cool the gas.

Binary fraction: $\sim 35\%$

Modern IMF: peak at $\log m \approx -1$

Primordial IMF: wide range of masses, cumilative IMF is logarithmically flat (thus top-heavy)

Present-day IMF:

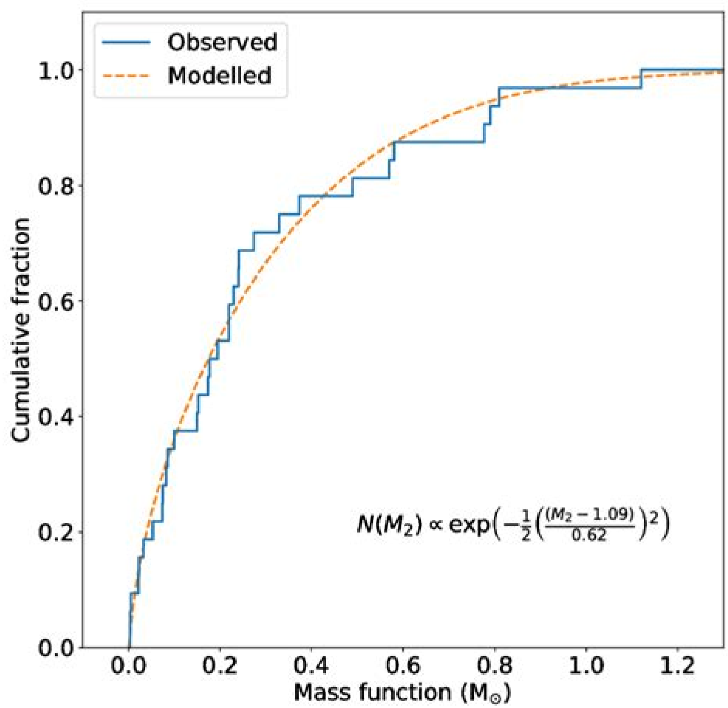

Figure 4

Primordial IMF:

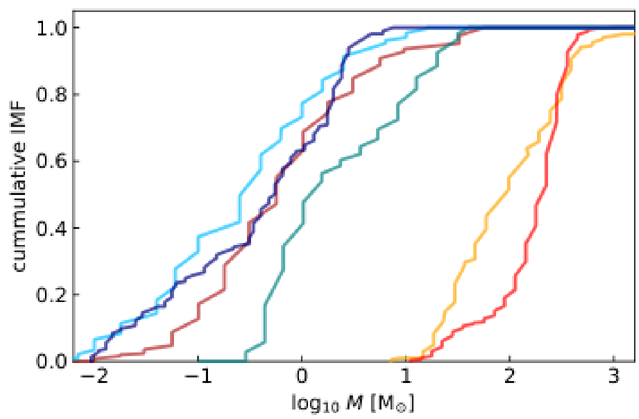

Figure 5

8 Fate

- Below $9M_{\odot}$: white dwarf, supported by electrong degeneracy pressure.
- $9M_{\odot} < M < 10M_{\odot}$, degenerate O-Ne cores that collapse
- $10M_{\odot} < M < 25M_{\odot}$, Fe cores that collapse; core-collapse supernova, NS remnant
- $25M_{\odot} < M < 40M_{\odot}$, Fe cores that collapse; core-collapse supernova, BH remnant (inner envelope falls back onto NS)
- $40M_{\odot} < M < 70 - 100M_{\odot}$, Fe cores that collapse; no supernova, BH remnant (PPISN?)
- $70 - 100M_{\odot} < M < 140M_{\odot}$, pulsational pair-instability SN (production of $e - e^{+}$ pair, triggers dynamical implosion of the core (in series), ejects mass until it's lower mass
- $140M_{\odot} < M < 260M_{\odot}$, pair-instability SN, no remnant
- $260M_{\odot} < M$, pair-instability SN, BH remnant

9 Feedback

9.1 Photoionization

Ionizing photons produced by a masive PopIII star are readily absorbed in the surrounding gas, leading to the formation of an HII region (ionized H). It initially expands in the bipolar region. Can disperse the cloud, inhibiting SF. Also, the shockwave of HII might increase the density, triggering SF

9.2 Photodissociation

Pop III stars emit many LW H_2 photons, triggering photodissociation

9.3 X-rays

Because of fragmentation, many Pop III stars form in tight binaries, which can become high-mass X-ray binaries (HMXBs)

9.4 Outflows

Outflows depend on initial magnetic fields. Outflows become important if the initial magnetic field is non-negligible. Simulations show that B-fields produced by the stars themselves with the dynamo effect do not seem to launch jets or outflows as one might expect.

9.5 Stellar winds

In Pop I stars, winds are driven by absorption of stellar photons by the lines of highly ionized metals. So the winds for Pop III stars are very weak and thus unimportant.

10 Transition to Pop II

Metallicity enchances the fragmentation of the gas and reduces the rate at which the fragmentas accrete adiitional material.

This leads to IMF dominated by low-mass stars, like present-day.

11 Footnote